

每日一題37

單元7 三角比-餘弦定理

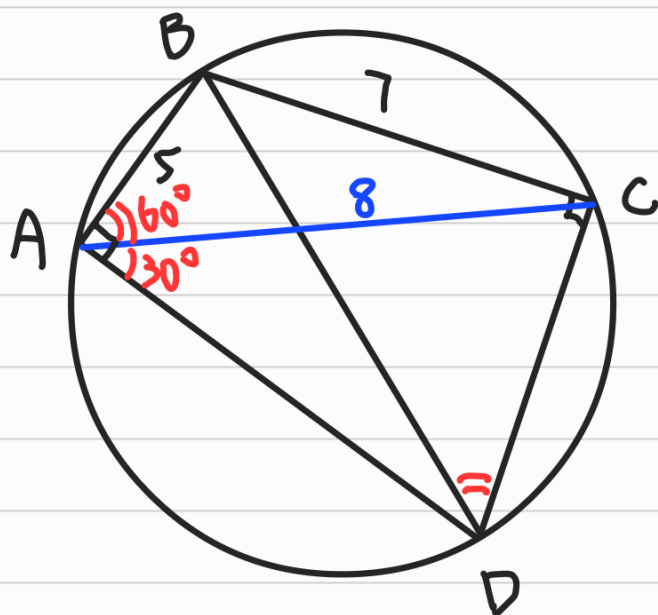
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114南一第=次模考 #16

圓內接四邊形 $ABCD$ 中， $\overline{AB} = 5$ 、 $\overline{BC} = 7$ 、 $\overline{AC} = 8$ 且 $\triangle DBC$ 為直角三角形，則

$$\overline{DA} = \frac{\sqrt{(16-1)(16-2)(16-3)(16-4)}}{3} \quad \text{。 (化為最簡分數)}$$

<Sol>



$$\begin{array}{r} \sqrt{148} \\ \sqrt{74} \\ 37 \end{array}$$

$$\because \angle BAC, \angle BDC \text{ 對 } \widehat{BC} \therefore \angle BAC = \angle BDC = \frac{1}{2} \widehat{BC}$$

$$\Rightarrow \cos \angle BAC = \cos \angle BDC = \frac{5^2 + 8^2 - 7^2}{2 \cdot 5 \cdot 8} = \frac{1}{2} = \cos 60^\circ$$

$$\overline{BD} = \frac{7}{\sin 60^\circ} = \frac{14}{\sqrt{3}}$$

$$\overline{CD}^2 = \left(\frac{14}{\sqrt{3}}\right)^2 - 7^2 = \frac{196}{3} - 49 = \frac{49}{3} \Rightarrow \overline{CD} = \frac{7}{\sqrt{3}}$$

$$\Rightarrow \left(\frac{7}{\sqrt{3}}\right)^2 = 8^2 + \overline{AD}^2 - 2 \cdot 8 \cdot \overline{AD} \cos 30^\circ$$

$$\Rightarrow \frac{49}{3} = 64 + \overline{AD}^2 - 8\sqrt{3}\overline{AD} \Rightarrow 49 = 192 + 3\overline{AD}^2 - 24\sqrt{3}\overline{AD}$$

$$\Rightarrow 3\overline{AD}^2 - 24\sqrt{3}\overline{AD} + 143 = 0$$

$$\Rightarrow (\sqrt{3}\overline{AD} - 11)(\sqrt{3}\overline{AD} - 13) = 0 \Rightarrow \overline{AD} = \frac{11}{\sqrt{3}} \vee \frac{13}{\sqrt{3}} \text{ (不合)}$$

$$\therefore \overline{AD} = \frac{11}{\sqrt{3}} = \frac{11\sqrt{3}}{3} *$$